

22

2026

WEEK 52

DECEMBER

TUESDAY

ديسمبر

الثلاثاء

٢٢

١٣ كيهك ١٧٤٣ ق

١٣ رجب ١٤٤٨ هـ

Abdelkhmon Ahmed KafSheen

Inputs

Outputs

a	b	Cin	Sum	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

$$Sum = a \bar{b} \bar{C}_{in} + a \bar{b} C_{in} + ab \bar{C}_{in} + ab C_{in}$$

evening

$$C_{out} = a \bar{b} C_{in} + ab \bar{C}_{in} + ab C_{in} + ab C_{in}$$

$$= C_{in} (a \bar{b} + ab) + C_{in} (a \bar{b} + ab)$$

$$C_{in} (a \oplus b) + C_{in} (a \oplus b)$$

$$= a \oplus b \oplus C_{in}$$

notes

21

2026

WEEK 52

DECEMBER

MONDAY

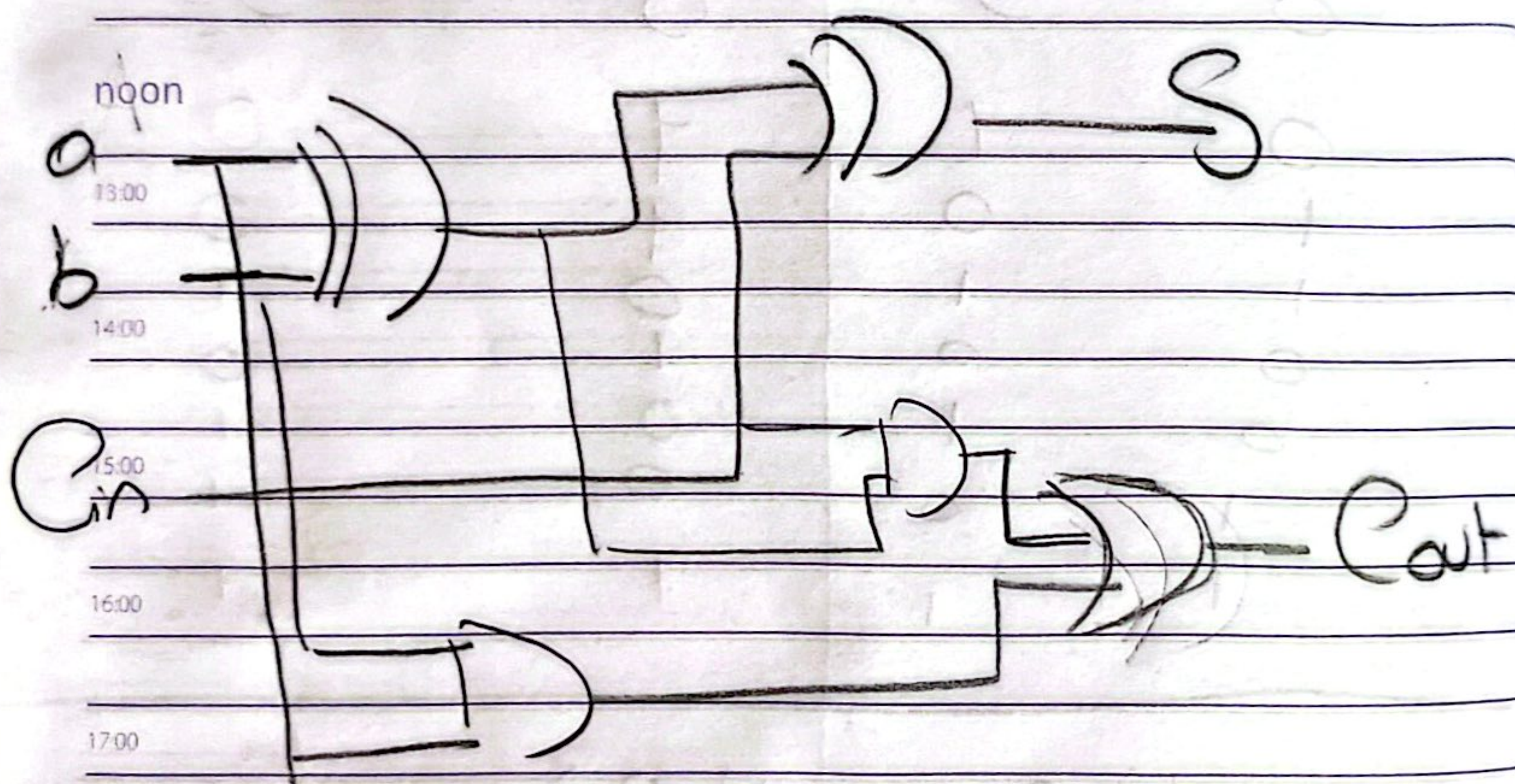
ديسمبر

الاثنين

١٢ كيهك ١٧٤٣ ق

١٤٤٨ هـ

$$\begin{aligned}
 C_{out} &= \bar{a}bC_{in} + a\bar{b}C_{in} + abC_{in} + ab \\
 &= C_{in}(\bar{a}b + a\bar{b}) + ab(C_{in} + C_{in}) \\
 &= C_{in}(a \oplus b) + ab \\
 &= ab + C_{in}(a \oplus b)
 \end{aligned}$$



evening

19:00

20:00

21:00

notes